Bitan Mukherjee

Final Year, B.E. Chemical Engineering
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EDUCATION

Jadavpur University **2023–2027**
Bachelor of Engineering in Chemical Engineering **CGPA: 9.1/10 | Departmental Rank 3**

- **Relevant Coursework:** Process Dynamics and Control, Transport Phenomena, Fluid Mechanics, Chemical Engineering Thermodynamics, Heat and Mass Transfer, Chemical Reaction Engineering, Numerical Methods, Separation Processes.
- **Technical Interests:** Process Modeling & Optimization, Sustainable Energy Systems, Colloidal Stability.

WORK EXPERIENCE

- **SET Intern | AB InBev India | Internship Completion Letter** **May'26–Jul'26**
  - Selected as one of **only 2 students out of 1000+ applicants** for Supply Excellence Program, designed to identify future leaders in manufacturing and supply by AB InBev India, home to **Budweiser, Corona Extra, and Hoegaarden**.
  - Collaborated cross-functionally across functions to understand operational workflows and **develop engineering solutions** for process improvement in a world-class manufacturing facility.
- **Research Assistant | Izmir Institute of Technology, Türkiye | LOR** **Mar'25–Nov'25**
  - Developed and implemented **Physics-Informed Neural Networks (PINNs) for advanced population balance modeling and conservatively perturbed equilibrium analysis**, improving predictive accuracy in chemical process simulations.
  - Developed MATLAB heat and fluid flow models and co-created course materials for CHE330 (Mathematical Modeling), validating model predictions against experimental benchmark **within 5% deviation**.
- **Summer Internship | Indian Institute Of Science, Bengaluru | LOR** **Jun'25–Jul'25**
  - Selected by NPTEL, Ministry of Education, for a fully-funded Summer Research Internship at the Kotak IISc AI-ML Centre, IISc Bengaluru, awarded to the **top 1 % of students across the nation for exceptional academic performance**.
  - Investigated advanced graph optimization techniques based on Minimum Spanning Trees, formulating **efficient algorithms for dynamic edge updates** and constrained network optimization with complexity analysis for large-scale engineering systems.
- **Remote Research Intern | IIM Ranchi | LOR** **Dec'24–Feb'25**
  - Conducted research on the Topic of "Spatio-Temporal Analysis and Prediction of Cybersecurity Risk Patterns **via LSTM and Graph Neural Networks approach**" under a Data Analytics project.
  - Developed a spatio-temporal forecasting framework using LSTM and Graph Neural Networks, **processing 40,000+ cybersecurity attack records** through feature engineering and state-level graph modeling for cybersecurity risk prediction.

PROJECTS

- **Renewable-Integrated PEM Electrolyzer System Design & Optimization | Explorer** **May'26**
  - Developed a **physics-constrained renewable hydrogen system optimization framework** integrating PEM electrochemistry, thermal dynamics, renewable power availability to predict system efficiency under real-world operating constraints.
  - Implemented a multi-parameter framework enabling **real-time evaluation of solar oversizing, electrolyzer sizing, stack scaling, and thermal management trade-offs** for renewable hydrogen systems.
- **PipeFlow Analyzer and Pump Selection Tool | MATLAB | Github** **Oct'25–Nov'25**
  - Built a MATLAB toolkit for pipe-flow analysis and automated pump selection,using Moody-chart friction factors and Darcy-Weisbach losses to match system **across 10 pump performance curves and validated accuracy within ±2% tolerance**.
  - Implemented system–pump curve modeling to automatically **identify the operating point from a 10-pump database (0.75–7.5kW), estimating efficiency (70–84%) and power requirements** with full visualization support.
- **Shell-and-Tube Heat Exchanger Designer | MATLAB | Github** **Nov'25–Dec'25**
  - Developed an end-to-end shell-and-tube heat exchanger design, sizing and optimization tool, **achieving ≤1% convergence accuracy using iterative U-value refinement** with Sieder–Tate and Kern correlations across 6 TEMA configurations.

RESEARCH PUBLICATIONS

Co-author: A. Dutta, **B.Mukherjee**, S.A.Hosen, M. Turan, D. Constales, and G. Yablonsky, "**A physics-informed neural network (PINN) approach to over-equilibrium dynamics in conservatively perturbed linear equilibrium systems**," *Entropy*, vol. 28, no. 1, p. 9, 2026. Available: `https://doi.org/10.3390/e28010009`

SKILLS & CERTIFICATIONS

- **Software Tools:** Python, MATLAB, DWSIM, OpenFOAM, AutoCAD, TensorFlow, PyTorch, NumPy, LaTeX, MS Office Suite (Word, Excel, Powerpoint)
- **Certifications:** **Machine Learning Specialization** | Stanford Online | `Credential`
  Introduction to Graph Algorithms | IISc Bengaluru | `Credential`
  Python for Data Science | IIT Madras | `Credential`